PHY 338k Lab Report 3

Lily Nguyen with Anna Grove

October 9, 2025

1 Introduction

Labs 7–8 of PHY 338k: Electronic Techniques explored further behavior and applications of operational amplifiers. Lab 7 focused on open-loop operation, inverting and summing amplifiers, and negative feedback controls. Lab 8 focused on measurement of input offset voltage, op-amp integrator function, and a battery-powered photodiode amplifier.

2 Lab 7: Operational Amplifiers I

2.1 Open-loop op-amp, comparator

We built the circuit in Figure VII-3 and found that it functions as an open-loop op-amp comparator. The input voltage $V_{\text{in}}(t)$ is a small-amplitude sine wave centered about zero and the output voltage $V_{\text{out}}(t)$ switches sharply between the maximum and minimum values where the input crosses zero.

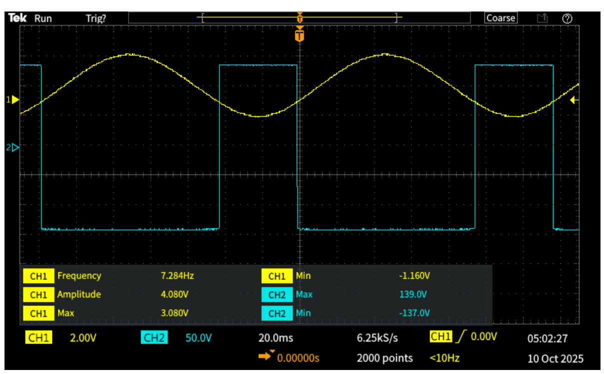


Figure 1: Oscilloscope display showing the open-loop op-amp acting as a comparator. CH1 is $V_{\text{in}}(t)$ and CH2 is $V_{\text{out}}(t)$.

The linear gain region of the op-amp (the slope near $V_{\text{in}} = 0$) is small and does not play a role in this circuit because the op-amp's open-loop gain is very large.

Even a tiny input difference would drive the output to the supply rails, so the circuit would saturate immediately rather than operate linearly. Thus, this op-amp acts as a comparator rather than an amplifier.

We measured the output voltages to be $V_{\mathrm{OM}}^{+} = 13.9\mathrm{V}$ and $V_{\mathrm{OM}}^{-} = -13.7\mathrm{V}$, which are both slightly less than the $\pm 15\mathrm{V}$ supply rails. According to the LF411 datasheet, the typical maximum output swing under a $10\mathrm{k}\Omega$ load is $\pm 13.5\mathrm{V}$. Our values were consistent with the specifications, so we confirm the op-amp's performs as expected.

Next, we changed the input to be a 4.32V square wave and measured the slew rate of the op-amp using the rise time. The slew rate is the fastest rate at which its output voltage can change.

$$\text{Slew rate} = \frac{\Delta V}{\Delta t} = \frac{27.6\mathrm{V}}{2.1\mu\mathrm{s}} \approx 13.1\mathrm{V}/\mu\mathrm{s} \tag{1}$$

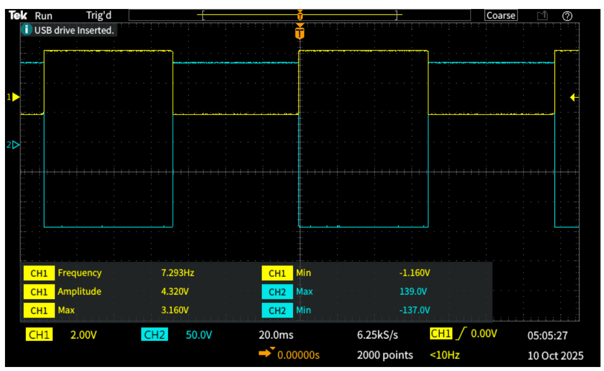


Figure 2: Oscilloscope display of the input voltage (CH1) and output voltage (CH2) for an open-loop op-amp comparato with a square wave input.

Our measured slew rate of approximately $13.1\mathrm{V}/\mu\mathrm{s}$ is close to the LF411 datasheet, which gives a typical slew rate of $13\mathrm{V}/\mu\mathrm{s}$.

2.2 Inverting amplifier

Next, we built the inverting amplifier circuit from Figure VII-4 using an LF411 op-amp. For a gain of -10, we used $R_{\mathrm{in}} = 9.16\mathrm{k}\Omega$ and $R_f = 102.2\mathrm{k}\Omega$. For a gain of -100, we used $R_{\mathrm{in}} = 5.2\mathrm{k}\Omega$ and $R_f = 556\mathrm{k}\Omega$. For both setups, we used a grounding resistor $R_g = 1.088\mathrm{k}\Omega$.

We applied a sine wave input from 10Hz to 1MHz and measured the input and output voltages on the scope to calculate the gain at each frequency. Figure 3 shows the gain vs. frequency on a log-log scale for both setups.

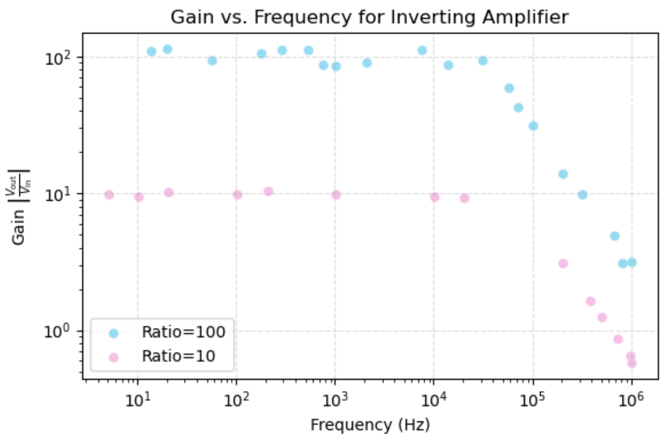


Figure 3: Plot of gain vs. frequency for resistance ratios of 10 and 100 on a log-log scale.

Fromm the roll-off points, we can estimate that the open-loop gain equals the closed-loop gain at the start of each drop. This would mean the open-loop gain is about 10^5 at 30kHz and 10^4 at 300kHz. From these two points, we estimate that the open-loop gain decreases approximately to $\frac{1}{f}$, This is consistent with a gain-bandwidth product on the order of 10^6Hz, which is expected for the LF411 op-amp.

Assuming we have an ideal-opamp (no input current, $V_+ = V_-$), the inverting intput is at virtual ground. When we apply Ohm's law:

$$I_{\text{in}} = \frac{V_{\text{in}}}{R_{\text{in}}} \tag{2}$$

$$I_f = \frac{-V_{\text{out}}}{R_f} \tag{3}$$

$$\tag{4}$$

setting $I_{\text{in}} = I_f$:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = \frac{-R_f}{R_{\text{in}}} \tag{5}$$

which shows that the gain depends only on the resistor ratio and includes a 180° phase shift.

2.3 Summing amplifier

Finally, we built a summing amplifier using the standard inverting amplifier configuation where the output voltage is:

$$V_{\text{out}} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right) \tag{6}$$

We kept both input voltages constant at $V_1 = V_2 = +5\text{V}$ and varied the the resistor vaues to observe out V_{out} depends on R_1, R_2, and R_f. Our measured values are summarized in Table 1 below.

Table 1: Measured output voltage V_{out} for various resistor combinations

R_1 (kΩ)	R_2 (kΩ)	R_f (kΩ)	V_{out} (V)
10.0	10.0	10.0	-8.75
5.34	10.0	10.0	-13.03
20.1	10.0	10.0	-6.75
10.0	5.34	10.0	-12.81
10.0	20.1	10.0	-6.53
10.0	10.0	5.34	-5.40
10.0	10.0	20.1	-13.55

Our results were consistent with expected behavior. For example, increasing R_f increased the magnitude of V_{out}, whereas increasing either R_1 or R_2 decreased their respective contribution of V_1 or V_2 to the output.

The general formula for an inverting summing amplifier with N inputs is:

$$V_{\text{out}} = -R_f \sum_{i=1}^{N} \frac{V_i}{R_i}, \tag{7}$$

which allows weighted contributions from each input voltage based on its resistor value.

3 Lab 8: Operational Amplifiers II

3.1 Op-amp input offset

We built an inverting amplifier with a gain $G = -1000$ from Figure VIII-2 to measure the input offset voltage V_{OS} of various op-amps. Since we grounded the input of the circuit ($V_{\text{in}} = 0$), the output voltage we measured was entirely due to the internal input offset voltage of the op-amp. Using the gain formula $V_{\text{out}} = G \cdot V_{\text{in}}$, we substituted $V_{\text{in}} = V_{\text{OS}}$ to get:

$$V_{\text{OS}} = \frac{V_{\text{out}}}{G} \tag{8}$$

Table 2 summarizes our data and calculations.

Op-Amp Type	Measured V_{out} (V)	Calculated V_{OS} (mV)
LF411	0.497 0.393 1.124	0.497 0.393 1.124
LM741	0.850 0.480 1.110	0.850 0.480 1.110
OPA227	0.163 0.145 0.147	0.163 0.145 0.147
LTC1150	0.153 0.153 0.159	0.153 0.153 0.159

Table 2: Measured output voltages and computed input offset voltages for various op-amps.

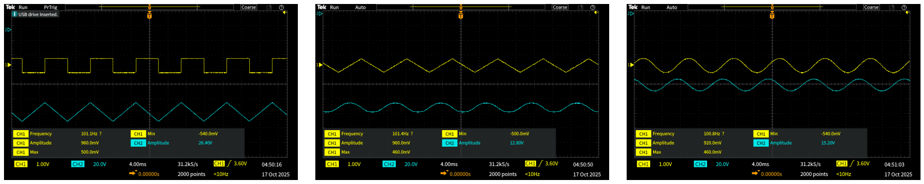
The LF411 op-amps had input offset voltages ranging from 0.393mV to 1.124mV, which were within the datasheet's typical value of 0.8mV and below the maximum of 2mV with variation among individual devices. The LM741 op-amps had offsets between 0.480mV and 1.110mV, which were also within the expected datasheet range (typical 1mV, maximum 5mV).

The OPA227 and LTC1150 are designed for extremely low offset voltages (10µV and 0.5µV typical), whereas our measured values were around 0.15mV. This is much higher than expected and likely due to limitations in our setup, such as thermoelectric effects or noise. Thermoelectric potentials appear at junctions between different metals when a temperature gradient is present and produces small voltage differences that are amplified significantly in high-gain circuits such as this one.

Overall, our measurements align with expected values for the LF411 and LM741, but not so much for the OPA227 and LTC1150 which had magnitudes lower expected values for the offset input voltage.

3.2 Op-amp integrator

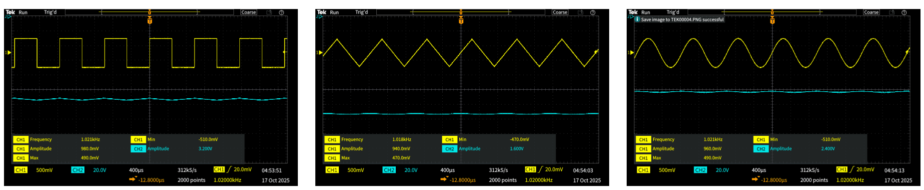
We built the damped op-amp integrator circuit from Figure VIII-5 an observed its response to square, triangle, and sine wave inputs at 100Hz and 1kHz. We initially left R_2 in place and kept the switch open for this setup.



(a) Square wave input (b) Triangle wave input (c) Sine wave input

Figure 4: Op-amp integrator waveforms for square, triangle, and sine inputs at $f = 100\text{Hz}$ with V_{in} (CH1) and V_{out} (CH2).

At 100Hz, the outputs behaved as expected for an ideal integrator. The square wave produced a triangular output, the triangle wave produced a parabolic output, and the sine wave showed a 90° phase shift resembling a cosine. At 1kHz, the output amplitudes decreased, which highlights the frequency-dependent gain of the integrator.



(a) Square wave input (b) Triangle wave input (c) Sine wave input

Figure 5: Op-amp integrator waveforms for square, triangle, and sine inputs at $f = 1\text{kHz}$ with V_{in} (CH1) and V_{out} (CH2).

When we removed R_2, the output no longer leveled off cleanly in response to the square wave. The tops and bottoms of the waveform became slanted rather than flat, which indicates the op-amp continued integrating any small DC offset. Originally, R_2 was included to limit DC gain and stabilize the output, so our observations make sense.

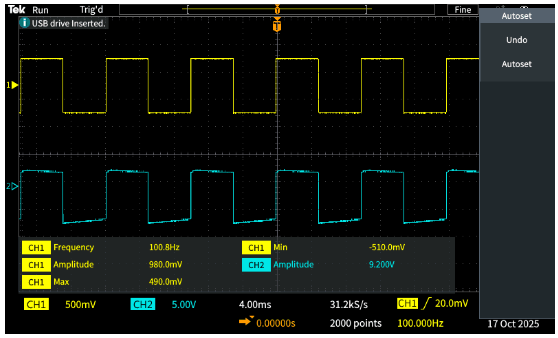


Figure 6: Op-amp integrator waveform with R_2 removed. V_{in} is CH1 and V_{out} is CH2.

With a 10kHz, 0.74V square wave input, we found that closing the switch caused the output to shift slightly upward and the amplitude to drop by about 0.2V. The switch effectively resets the integrator by discharging the capacitor, which clamps the integrator to a baseline voltage.

The passive RC circuit from Lab 2 acts as a low-pass filter, but it can behave as an integrator when the input frequency is much higher than the circuit's cutoff frequency ($f_c = \frac{1}{2\pi RC}$). However, since it lacks the feedback mechanism present in the op-amp circuit, it only approximates the integration and only works under specific conditions.

The op-amp integrator from this lab uses negative feedback to maintain a virtual ground at the inverting input, which allows it to integrate more accurately over a wider range of frequencies, including lower ones. In contrast, if we operate the passive RC circuit below its cutoff frequency, it would behave more like a voltage divider and its output would not represent the integral of the input.

Although both circuits have the same basic layout, they can be represented as low-pass filters. The op-amp integrator behaves like a true integrator when the input frequency is well below the cutoff freuqency $f_c = \frac{1}{2\pi RC}$. As the frequency approaches or exceeeds the cutoff, the circuit begins to attenuate the high-frequency components rather than integrate, so its behavior would better be described as that of a low-pass filter.

3.3 Battery-operated photodiode circuit

3.3.1 Battery-powered photodiode amplifier

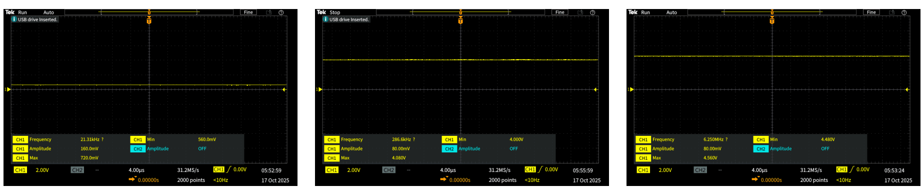
Lastly, we built the battery-powered photodiode amplifier from Figure VIII-7. We used a feedback resistor of $R = 1.17\text{M}\Omega$, which makes the transimpedance gain of this circuit:

$$\text{Gain} = \frac{V_{\text{out}}}{I_{\text{in}}} = R = 1.17 \times 10^6 \Omega \tag{9}$$

We tested the circuit under four lighting conditions and recorded the output voltage using the oscilloscope:

1. Light blocked: $V_{\text{out}} \approx 0.72\text{V}$ (used as baseline)
2. Room light only: $V_{\text{out}} \approx 4.08\text{V}$
3. Phone light at a distance: $V_{\text{out}} \approx 4.25\text{V}$
4. Phone light up close: $V_{\text{out}} \approx 4.56\text{V}$

As expected, the output voltage increased with light intensity. Our circuit remained stable and did not oscillate, so there was no need to add additional components such as a compensating capacitor.



(a) Light blocked (b) Room light only (c) Phone flashlight (close)

Figure 7: Photodiode amplifier output under various lighting conditions. As expected, the output voltage increases with light intensity.

Using the photodiode's responsivity value of approximately 0.5A/W, we estimated the minimum light power we can detect with our photodiode and circuit. The difference in $V_{\text{out, min}}$ from the blocked light to the room light case was roughly:

$$\Delta V = 4.00 - 0.56 = 3.44\text{V} \tag{10}$$

which corresponds to a photocurrent of:

$$I = \frac{\Delta V}{R} = \frac{3.44}{1.17 \times 10^6} \approx 2.94\mu\text{A} \tag{11}$$

Using the responsivity:

$$P = \frac{I}{0.5} \approx 5.88\mu\text{W} \tag{12}$$

So, the minimum light power we could detect was approximately 5.9µW.

For the brightest case (phone light up close, $V_{\text{out}} \approx 4.56\text{V}$), assuming the baseline is still approximately 0.56V, the change is:

$$\Delta V = 4.56 - 0.56 = 4.00\text{V} \tag{13}$$

$$I = \frac{4.00}{1.17 \times 10^6} \approx 3.42\mu\text{A} \tag{14}$$

$$P = \frac{I}{0.5} \approx 6.84\mu\text{W} \tag{15}$$

So, the maximum light power detected was approximately 6.8μW.

These values are reasonable for a small silicon photodiode in a low-power indoor setup. The amplifier's high gain enabled it to convert the small changes in light intensity to measurable voltage changes.

Conclusion

Labs 7–8 gave us practical insight into the versatility of op-amps in analog circuit design. We were able to demonstrate their use in signal processing and sensing along with a variety of other applications. Overall, these experiments reinforced the central role of op-amps as adaptable components in modern electronics.

Acknowledgements

We would like to thank our laboratory instructor, Dr. Heinzen, and our teaching assistant, Kyle Gable, for their guidance and support through these labs.Their assistance and feedback were invaluable in helping us understand the experimental procedures and circuit concepts. We also appreciate the department's laboratory facilities and equipment, which made these experiments possible.